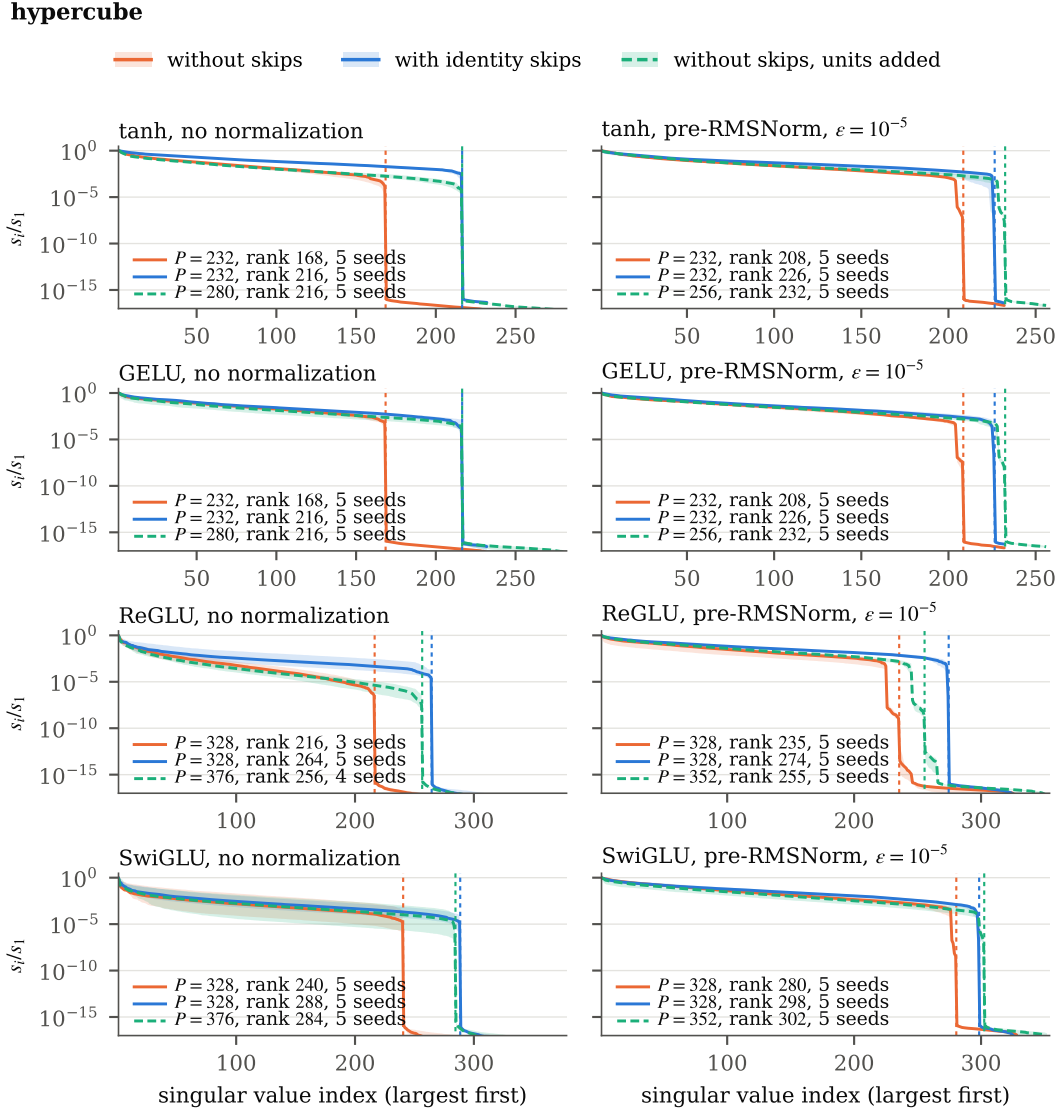

hypercube
without skips
with identity skips
without skips, units added
tanh, no normalization
tanh, pre-RMSNorm, $\varepsilon = 10^{-5}$
GELU, no normalization
GELU, pre-RMSNorm, $\varepsilon = 10^{-5}$
ReGLU, no normalization
ReGLU, pre-RMSNorm, $\varepsilon = 10^{-5}$
SwiGLU, no normalization
SwiGLU, pre-RMSNorm, $\varepsilon = 10^{-5}$
s_i/s_1
$P = 232$, rank 168, 5 seeds
$P = 232$, rank 216, 5 seeds
$P = 280$, rank 216, 5 seeds
$P = 232$, rank 208, 5 seeds
$P = 232$, rank 226, 5 seeds
$P = 256$, rank 232, 5 seeds
$P = 328$, rank 216, 3 seeds
$P = 328$, rank 264, 5 seeds
$P = 376$, rank 256, 4 seeds
$P = 328$, rank 235, 5 seeds
$P = 328$, rank 274, 5 seeds
$P = 352$, rank 255, 5 seeds
$P = 328$, rank 240, 5 seeds
$P = 328$, rank 288, 5 seeds
$P = 376$, rank 284, 5 seeds
$P = 328$, rank 280, 5 seeds
$P = 328$, rank 298, 5 seeds
$P = 352$, rank 302, 5 seeds
singular value index (largest first)
singular value index (largest first)